Learner study · facilitator's session script

For the team member running sessions with 4–5 adult Hindi speakers who do not know Telugu. Website: **https://favianaNoronha.github.io/LMA---Reverse-learning-Tutor/ → Research team** page for the study tools.

Before the first session

- Ethics approval received; consent form filled in with the reference and contacts (two copies per person).
- **One browser profile per participant** on the study laptop (Chrome: profile icon → Add; name it P01, P02, ...). The website keeps all progress in the browser, so each participant must always use their own profile on the same laptop, and nobody clears browser data during the study.
- In each profile: open the website → Research team → Study → type the code (P01...) → **Assign conditions**. This splits the 15 rules into three groups of 5: A (teach Raju), B (study examples), C (normal tutor), rotated across participants.
- Write down each person's condition order for practice (counterbalanced): P01 A→B→C, P02 B→C→A, P03 C→A→B, P04 A→C→B, P05 B→A→C.
- Plan the dates: Day 1, Day 2, Day 3, and the delayed test exactly 7 days after Day 3 (±1 day).

Session plan

DAY	STEPS	TIME
1	Welcome · consent · background questions · pre-test · letters tour · first practice block	45 min
2	Practice block	20 min
3	Practice block · post-test · feedback form	30 min
3 + 7 days	Delayed test · transfer test · short interview	20 min

Day 1

1. Welcome (2 min).

> *"Thanks for coming. We're testing a website that teaches basic Telugu to Hindi speakers. We're testing the website, not you, so there are no wrong answers about it. It takes about 45 minutes today and three shorter sessions later."*

2. Consent and background (8 min). Give the consent form; let them read; answer questions; both sign. They fill in the background questions with their code only. Check the exclusion line.

3. Pre-test (5 min). Research team → Study → Tests → **Pre-test** → Start test, then hand over the laptop.

> *"This is a short test before you learn anything. You'll see a sentence in English and Hindi. Type it in Telugu, or in Hindi letters, as best you can. If you don't know, press 'I don't know'. That's completely normal today."*

4. Letters tour (5 min). Open **Letters** and let them go through the three steps.

> *“Telugu letters are cousins of Hindi letters, in the same order. Have a look; you don't need to learn them all. In the lessons you can always type in Hindi letters.”*

5. Practice block (20 min), in their condition order. About 6–7 minutes per condition:

CONDITION	WHERE	WHAT TO SAY
A teach Raju	Research team → Study → **Practise A in Learn** (opens Lessons, limited to A rules)	“Learn the rule, then check Raju's sentences. If he's wrong, fix his sentence.”
B examples	Research team → Study → **Next example**, repeatedly	“Read each wrong sentence, the correction and the explanation. Take your time.”
C normal tutor	Research team → Study → **Next sentence**, type, **Check**	“Write the sentence in Telugu; the tutor corrects you.”

Aim for the **same number of items in each condition** per session (about 6). Note the counts on the record sheet.

Day 2

Practice block as on Day 1, same order, about 6 items per condition (20 min).

Day 3

Practice block (20 min). Then **Post-test** (Research team → Study → Tests → Post-test). Then the **Feedback** page (System Usability Scale + 3 questions). Stay out of view while they answer.

One week after Day 3

1. Delayed test, then **Transfer test** (new sentences). No practice before the tests.

> *“Same as the first test. Type the sentence in Telugu or Hindi letters. Don't worry if you've forgotten some.”*

2. Interview (5 min). Note answers in writing, no recording:

1. Which way of practising did you like most, and why?
2. Did correcting Raju feel like teaching, or like a quiz?
3. What was confusing?
4. Did the Hindi letters help, or get in the way?
5. Would you use something like this to learn Telugu? What would make you use it?

3. Export the data (do this at the end of *every* session too):

- Research team → Study → **Copy results as CSV** → paste into the team sheet *study_results* (one tab per participant).
- Research team → Demo controls → **Copy all feedback as CSV** → paste into *feedback*.
- Do **not** press Reset until all data is exported and checked.

Rules for the facilitator

- Never tell a participant whether an answer is right during practice or tests; the website does that during practice, and nobody does it during tests.

- Give the same help in every condition. The reading aids (Hindi letters, listen button, Hindi-letter typing) stay on for all three.
- Don't explain the rules yourself. If asked, say: “Use what's on the screen; you're doing fine.”
- Test answers are marked later by a Telugu teacher from **outside** the team who does not know which condition each rule was in.
- If something breaks, write down what happened and the time, and continue in the same browser profile.

Record sheet (one per participant)

CODE	ORDER	DAY 1 DATE · ITEMS A/B/C	DAY 2 DATE · ITEMS A/B/C	DAY 3 DATE · ITEMS A/B/C	DELAYED TEST DATE	DATA EXPORTED?

The paper (Section 8.3) describes this design: within-learner, three conditions on different rules by Latin square, matched exposure and feedback, delayed and transfer tests marked blind. With 4–5 learners, results are reported as effect sizes with confidence intervals, without significance claims.